

Bitwise differential trail through **QALQAN** found by SAT

CiVerLy

August 21, 2026

1 QALQAN

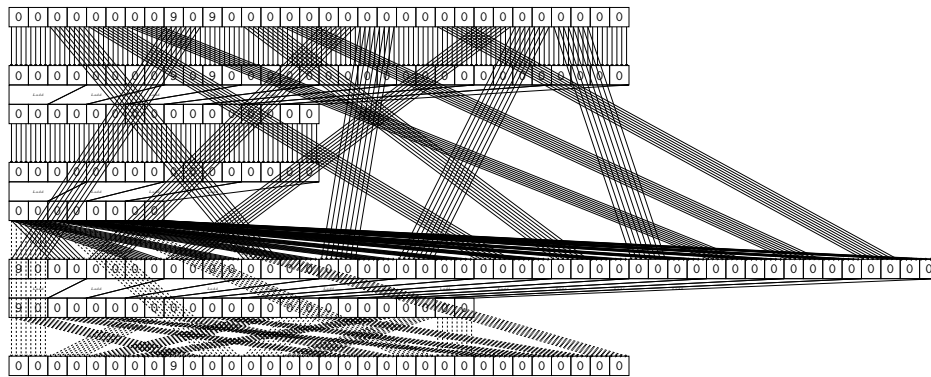
Maximal differential probability: 2^{-93} [illegible]

2 SBoxLayer

Maximal differential probability: $2^{-12.0}$ [illegible]

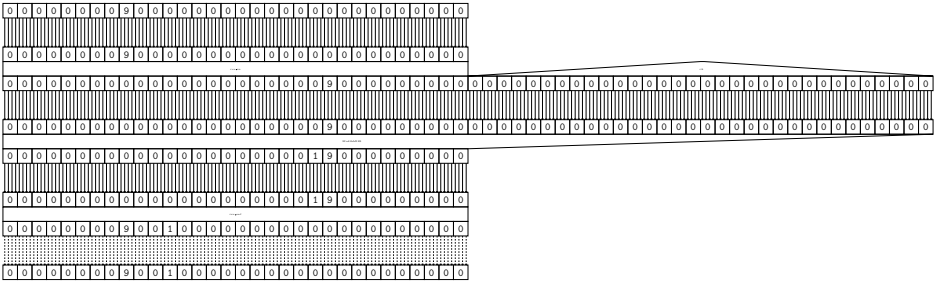
3 L

Maximal differential probability: $2^{-2.0}$



4 Add128

Maximal differential probability: $2^{-3.0}$



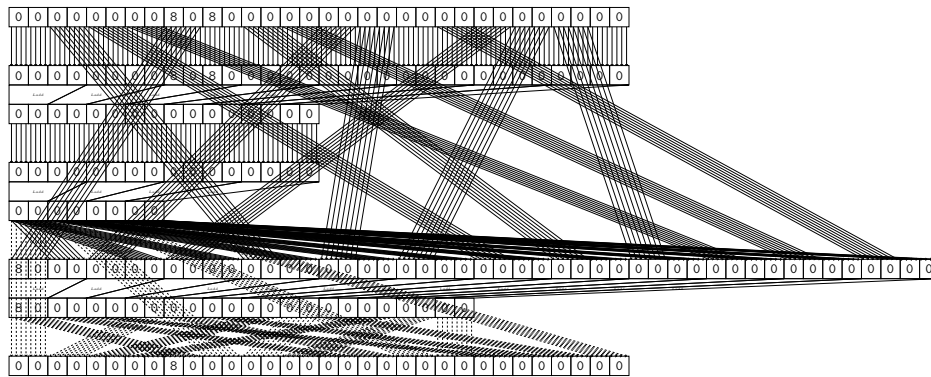
5 SBoxLayer

Maximal differential probability: $2^{-14.0}$

0	0	0	0	0	0	0	0	9	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	9	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	0	0	0	0	0	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

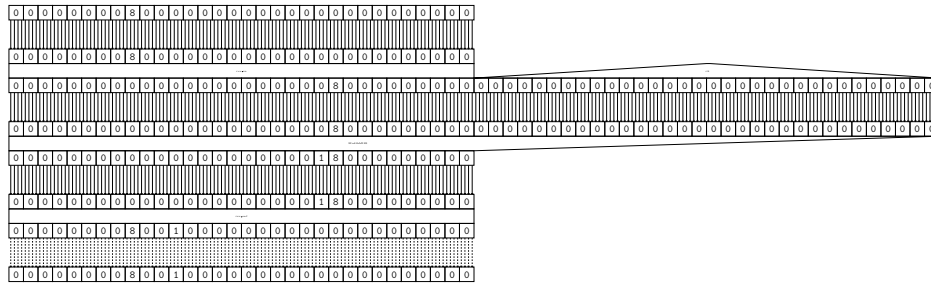
6 L

Maximal differential probability: $2^{-0.0}$



7 Add128

Maximal differential probability: $2^{-2.0}$



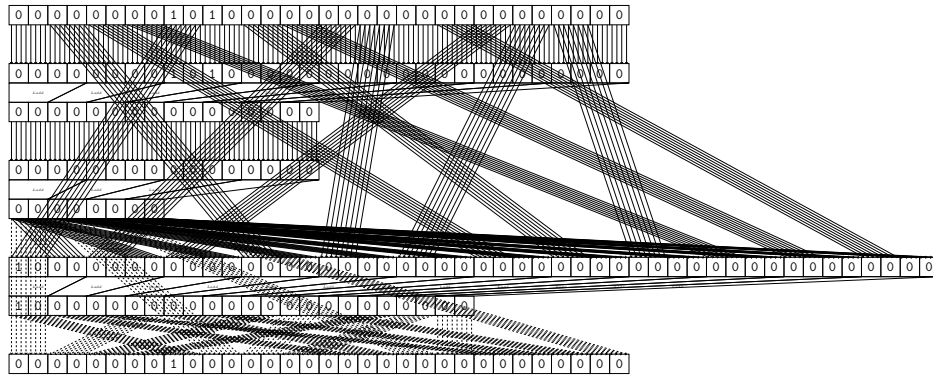
8 SBoxLayer

Maximal differential probability: $2^{-14.0}$

0	0	0	0	0	0	0	0	8	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

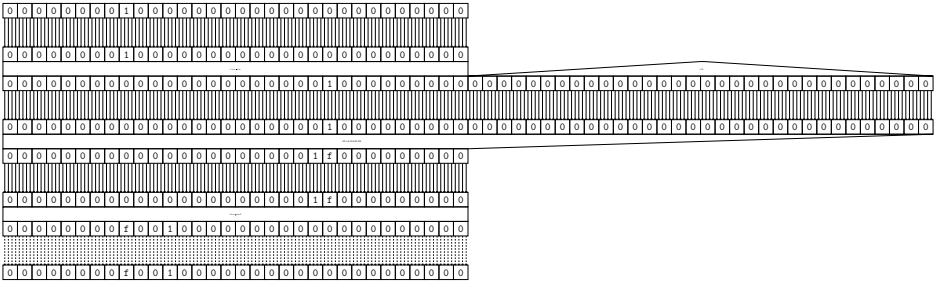
9 L

Maximal differential probability: $2^{-2.0}$



10 Add128

Maximal differential probability: $2^{-5.0}$

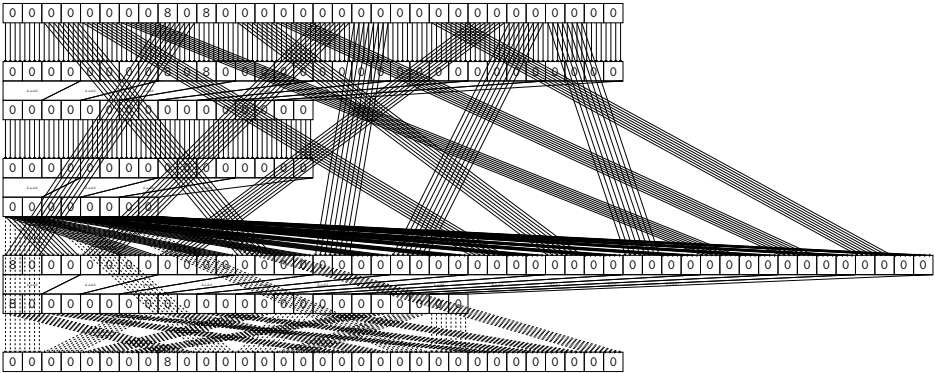


11 SBoxLayer

Maximal differential probability: $2^{-13.0}$ [illegible]

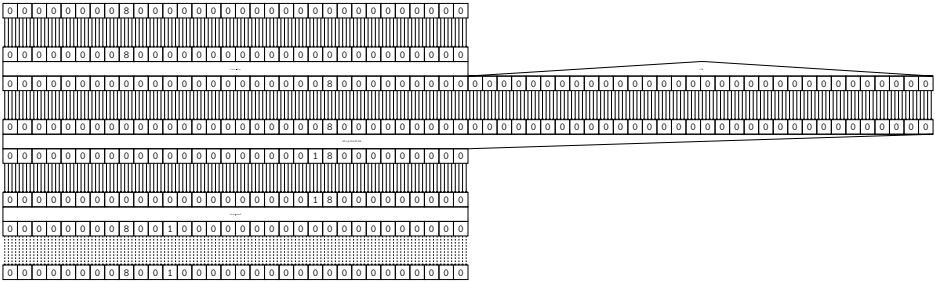
12 L

Maximal differential probability: $2^{-0.0}$



13 Add128

Maximal differential probability: $2^{-2.0}$



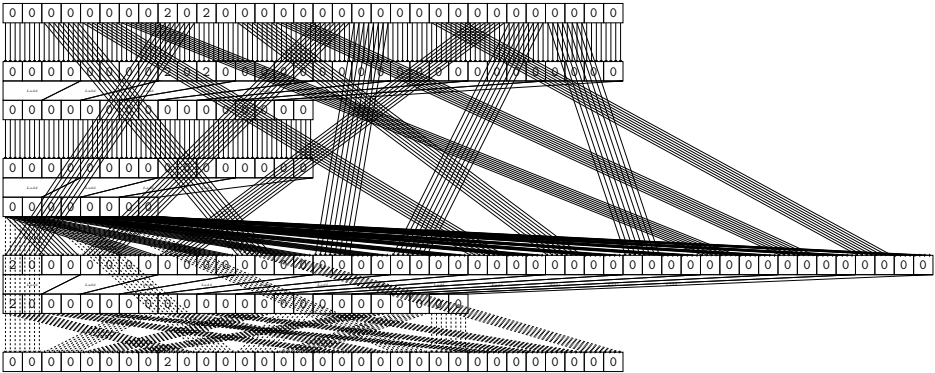
14 SBoxLayer

Maximal differential probability: $2^{-14.0}$

0	0	0	0	0	0	0	0	8	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000	0 0000
0	0	0	0	0	0	0	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

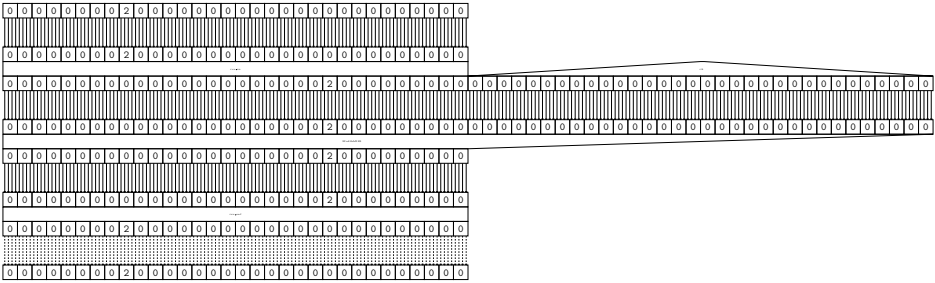
15 L

Maximal differential probability: $2^{-2.0}$



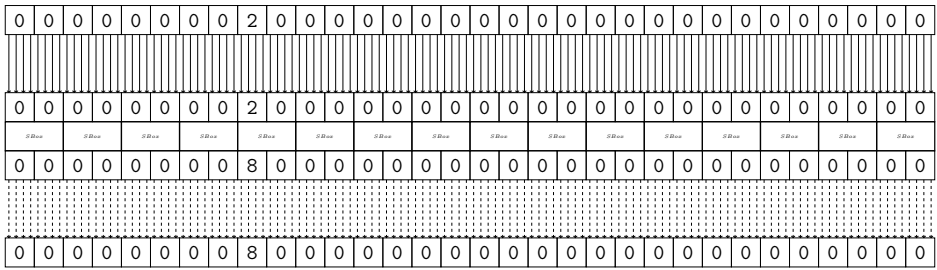
16 Add128

Maximal differential probability: $2^{-1.0}$



17 SBoxLayer

Maximal differential probability: $2^{-7.0}$



18 L

Maximal differential probability: $2^{-0.0}$

